Ramakrishna Saravanan

+1 224-382-4456 | ramakrishna1106s@gmail.com | Rama's portfolio | linkedin.com/in/ramakrishna | github.com/RickuSicku

Education

Northwestern University — Evanston, IL
Master of Science in Artificial Intelligence (GPA: 4.0) — *Sep 2025 – Dec 2026 (Expected)*

Indian Institute of Technology Patna — Patna, India
B.Tech in Engineering Physics (CPI: 8.03) — *July 2021 – May 2025*

Experience

Samsung Research Institute Bangalore (PRISM) — July 2024 – Mar 2025
Visual Intelligence Team, IIT Patna — *Bangalore, India*

- Designed an **on-device visual perception pipeline** for video understanding, focusing on **image quality assessment and frame selection** under mobile and embedded compute constraints.
- Optimized **GPU-accelerated inference pipelines** to achieve real-time performance (processing ~10% of video length), balancing **latency, memory efficiency, and deployability**.
- Technologies Used : *Python, TensorFlow, PyTorch, HuggingFace, OpenCV, OpenCLIP, Docker.*

SPARK Summer Research Internship, IIT Roorkee — May 2024 – July 2024
Summer Research Intern — *Roorkee, India*

- Built end-to-end **data preprocessing and modeling pipelines**, transforming raw imaging data into structured datasets suitable for predictive modeling.
- Fine-tuned and optimized a **U-Net–based image restoration and segmentation model**, achieving **97.3%** accuracy, with emphasis on **noise suppression and fine-detail preservation**.
- Technologies Used : *Python, Tensorflow, Pytorch, OpenCV, Scikit-learn, SAM2, Vision Transformers, Tkinter.*

Projects

Self-Directed RL & Logic Generation System — Jan 2026 – March 2026

- Engineered a **PPO-based RL agent** with an Actor-Critic CNN and dynamic action masking to autonomously solve Minesweeper with high win rates.
- Developed an **LLM-driven pipeline** using Azure OpenAI to convert unstructured rules into Event Calculus for automated reward design and reasoning.
- Technologies: *Python, PyTorch, Reinforcement Learning (PPO), Azure OpenAI, Streamlit.*

Youtube Video Analyzer — Jan 2026 – Mar 2026

- Built an **Google Extension** which works based on a **two-stage LLM pipeline** using Gemini + Search Grounding for zero-shot claim extraction, evidence verification, and confidence scoring.
- Engineered a **hybrid emotion-scoring system** combining VADER with LLM-based context windows to capture nuanced sentiment, rhetoric intensity, and claim certainty.
- Technologies: *Python, Google GenAI (Gemini), FastAPI, VADER.*

Enterprise RAG System for Technical Documentation — Oct 2025 – Dec 2025

- Architected a scalable **RAG framework** using LangChain and OpenAI LLMs to generate accurate, context-grounded answers for enterprise documentation queries.
- Built a high-performance **dense retrieval pipeline** with semantic chunking, HuggingFace embeddings, and FAISS-based ANN search for fast and relevant document retrieval.
- Technologies: *Python, LangChain, HuggingFace, FAISS, OpenAI, FastAPI.*

Technical Skills

Data Science & Analytics: Python, SQL, Pandas, Numpy, Scikit-learn
LLMs & Generative AI: Tensorflow, Pytorch, LLM, RAG, LangChain, OpenAI, NL2SQL, Agentic AI, Prompt Engineering, LLM Benchmarking & Evaluation, Text Embeddings, Vector Search.
Systems & Deployment: Docker, Kubernetes, Azure
Frontend: JavaScript, React, HTML/CSS